OSCE COACH

When Do We Actually Give Antibiotics?

The Therapeutic Guidelines decision tree for dental practice, in one place

Compiled from Therapeutic Guidelines: Oral and Dental
with page references throughout

Study aid for the ADC practical exam.
Not a prescribing reference.

TG's answer to "when do we give antibiotics" is scattered across about fifty pages — acute infections in one chapter, prophylaxis in another, necrotising gingivitis and peri-implantitis somewhere else again. In the exam it comes at you as a single question. This pulls the whole decision into one place, with page references back to TG so you can check anything that matters.

Antibiotics are not a substitute for dental treatment.

Drainage and source control are the treatment. Antibiotics are an adjunct, and only in specific situations. If you remember one sentence from this document, make it that one — it is the sentence TG itself keeps repeating, and it answers more exam questions than any drug name.

1. Acute odontogenic infection: the three tiers

Everything in acute infection follows from which of three tiers the patient is in. TG classifies by **facial swelling** and **severe local or systemic features** — not by how sore the tooth is. [Table 13.19, p126–127]

TIER	WHAT YOU SEE	WHAT YOU DO
Localised	Dental pain, often marked tenderness to percussion. Abscess or pus may be visible. **No facial swelling.** No severe features.	A dental procedure in the community. **No antibiotics** if that procedure happens within 24 hours.
Spreading, no severe features	**Facial swelling.** Dental or facial pain. Abscess or pus may be visible. No severe features.	Dental procedure **plus** oral antibiotics. This is the main indication for antibiotics.
Spreading with severe local or systemic features	See the red-flag list below.	**Urgent transfer to hospital** with an OMFS/anaesthetic/emergency facility. IV antibiotics and urgent surgery.

The red flags that make it tier three

Severe local features (airway risk): difficulty breathing · difficulty swallowing · swelling of the floor of the mouth · trismus · neck swelling · significant facial swelling and pain, especially with any of the above. Swelling of the upper third of the face occluding the eye is rare and may indicate cavernous sinus thrombosis.

Systemic features: pallor · sweating · tachycardia · temperature above 38°C or below 36°C.

Signs of sepsis in adults: impaired consciousness · respiratory rate 22+ · systolic BP under 90 mmHg · poor peripheral perfusion or mottled skin · acute drop in urine output.

While you wait for transfer: give oxygen by mask if they are struggling to breathe or look blue, let them adopt a position of comfort, and phone ahead to the receiving service both to alert them and to get advice on prehospital management.

TG's own caveat: acute odontogenic infections are a continuum. If you are unsure of the severity, seek expert advice — saying that out loud in a station is a strength, not a hedge.

2. The only three indications in acute infection

This is the list to be able to recite. [p127–128]

Antibiotic therapy is not a routine adjunct in managing acute odontogenic infections. Antibiotics are indicated:

- as an **adjunct to a dental procedure** in managing **spreading** odontogenic infections
- if a dental procedure to drain the infection **is not likely to take place within 24 hours**
- if a **fragment or root of an infected tooth is retained** during extraction and complete extraction is delayed

The trap: antibiotics are *not* indicated if a **non-infected** tooth fragment or root is retained during an extraction. Examiners like this one because it sounds like it should be a yes.

Spreading infection is the main indication, because of the risk of airway compromise or sepsis. Where treatment is delayed or a fragment is retained, the antibiotic is buying time — it aims to stop a localised infection becoming a spreading one.

Tell the patient explicitly that dental treatment is still essential as soon as possible **regardless of how much better they feel** on the antibiotic. TG calls this out specifically, and it is an easy communication mark.

Two situations that override the tiers

- **Profound immune compromise** — at risk of rapidly spreading infection even from a localised one. Urgent management, and make the antibiotic/timing decision **in consultation with their specialist or medical practitioner**.
- **Recurrent infection already treated with antibiotics but no dental treatment** — seek expert advice.

3. What to actually prescribe

For spreading odontogenic infection **without** severe features, managed in the community. [p132–134]

If the dental procedure happens within 24 hours — penicillin monotherapy

phenoxymethylpenicillin 500 mg (child: 12.5 mg/kg up to 500 mg) orally, 6-hourly for 5 days

OR amoxicillin 500 mg (child: 15 mg/kg up to 500 mg) orally, 8-hourly for 5 days

Phenoxymethylpenicillin is ***preferred*** *over amoxicillin because its spectrum is narrower. Both can be taken with or without food.*

If the procedure is delayed beyond 24 hours or incomplete — add anaerobic cover

metronidazole 400 mg (child: 10 mg/kg up to 400 mg) orally, 12-hourly for 5 days

PLUS phenoxymethylpenicillin 500 mg 6-hourly, OR amoxicillin 500 mg 8-hourly, for 5 days

OR as a single drug: amoxicillin + clavulanate 875 + 125 mg orally, 8-hourly for 5 days

Amoxicillin + clavulanate is an acceptable alternative if adherence to a two-drug regimen is unlikely, but it is ***broader spectrum and therefore not preferred*** *on stewardship grounds. Metronidazole* ***alone*** *is never right for odontogenic* *infection — no aerobic activity.*

Penicillin hypersensitivity

Non-severe reaction (urticaria, mild immediate rash, benign childhood rash, maculopapular rash): cephalexin 500 mg (child: 12.5 mg/kg up to 500 mg) orally, 6-hourly for 5 days

Severe reaction (anaphylaxis, airway compromise/angioedema, hypotension, collapse; or DRESS, SJS/TEN, severe blistering rash, organ involvement): clindamycin 300 mg (child: 10 mg/kg up to 300 mg) orally, 8-hourly for 5 days

Clindamycin — even short courses — raises the risk of *Clostridioides difficile* diarrhoea. Tell the patient: if diarrhoea starts, stop the clindamycin and contact you and their GP.

Both cephalexin and clindamycin are broader than penicillin, so they are alternatives rather than equals — stewardship again. And before you reach for either: ***verify the allergy****. TG devotes several pages to the fact that most reported penicillin allergy is not real.*

Follow-up

If the response to penicillin monotherapy is inadequate at **48 hours** after the dental procedure, that is the trigger to add metronidazole.

4. Surgical prophylaxis — almost never

Surgical antibiotic prophylaxis is rarely indicated for dental procedures.

This is the heading TG uses. In general dental practice the indications are rare enough that the safe default is no.

[Table 13.12, p76–77]

DOES NOT NEED SURGICAL PROPHYLAXIS	DOES NEED SURGICAL PROPHYLAXIS
third molar (wisdom tooth) surgery tooth extractions insertion or removal of dental implants, and peri-implant surgery periodontal surgery periapical surgery soft- and hard-tissue removal (biopsies, exostoses, alveoloplasty)	some OMFS procedures only: • insertion of prosthetic material — **except dental implants** • ORIF of mandibular or midfacial (Le Fort, zygomatic) fractures • intraoral bone grafting • orthognathic surgery • cleft lip and palate repair

Joint prostheses, breast implants, ventriculoperitoneal and ventriculo-atrial shunts, and deep brain stimulators do NOT change the advice. The hip replacement patient who has been told by someone to always take antibiotics before the dentist is a stock ADC station. The answer is no.

Do not use prophylactic antibiotics to prevent dry socket. Alveolar osteitis is premature clot lysis, not infection, and antibiotics do not prevent it.

Implants specifically: prophylaxis does not reduce postoperative infection. It is associated with a marginal 1–2% increase in implant retention — which is not an infection indication.

Before you even ask about prophylaxis, **treat existing infection** — an acute periapical infection or necrotising gingivitis is a treatment question, not a prophylaxis one.

5. Endocarditis prophylaxis — both boxes must tick

Prophylaxis needs a high-risk PROCEDURE and a high-risk CARDIAC CONDITION.

One without the other is not an indication. Candidates lose marks by remembering the valve list and forgetting that a non-invasive procedure needs nothing at all.

[Table 2.59, p82–83]

Procedures that qualify

Dental procedures **only those involving manipulation of gingival or periapical tissue, or perforation of the oral mucosa** — extractions, implant placement, biopsy, removal of soft tissue or bone, scaling (supra- and subgingival debridement), replanting avulsed teeth.

Cardiac conditions that qualify

- prosthetic cardiac valve, including transcatheter-implanted prosthesis or homograft
- prosthetic material used for cardiac valve repair (annuloplasty rings, chords)
- previous infective endocarditis
- ventricular assist devices
- congenital heart disease, **but only** unrepaired cyanotic defects (including palliative shunts and conduits), or repaired defects with residual defects at or adjacent to a prosthetic patch or device
- rheumatic heart disease

This one has changed. Endocarditis prophylaxis was previously recommended for rheumatic heart disease only in Aboriginal and Torres Strait Islander peoples, or considered for people at significant socioeconomic disadvantage. It now applies to **rheumatic heart disease in anyone** — RHD is an independent risk factor. If you learned the old rule, relearn this one.

Not indicated for mitral valve prolapse, septal defects, or cardiac implantable electronic devices (pacemakers, ICDs). Heart transplant patients who have developed cardiac valvulopathy may be high risk — consult their cardiologist.

The regimen

amoxicillin 2 g (child: 50 mg/kg up to 2 g) orally, **60 minutes before** the procedure

Non-severe penicillin reaction: cephalexin 2 g (child: 50 mg/kg up to 2 g) orally, 60 minutes before

Severe penicillin reaction: doxycycline 100 mg orally, or azithromycin 500 mg (child: 10 mg/kg up to 500 mg) orally, 60 minutes before

Clindamycin is no longer recommended for endocarditis prophylaxis — the rate of severe adverse reactions, mostly *C. difficile*, is higher than the alternatives. If your notes still say clindamycin, they are out of date.

Practical points on the alternatives: doxycycline can cause oesophageal ulceration — take with a large glass of water, stay upright 30 minutes. Azithromycin prolongs QT — check with the patient's doctor before prescribing. If oral is impossible, amoxicillin or ampicillin 2 g IM 30 minutes before, or IV within 60 minutes before (cefazolin 2 g for non-severe penicillin allergy).

The bit candidates skip

TG is blunt that prophylaxis is the *weaker* half of prevention. Everyday toothbrushing and flossing produce bacteraemia at a rate comparable to most dental procedures, and they happen far more often — and that bacteraemia is strongly associated with **poor oral hygiene and gingival disease**. So maintaining good oral health is likely more important than the prophylactic dose. Also advise against tattooing and piercing, especially tongue piercing, and investigate any unexplained fever.

6. Named conditions — yes, no, rarely

CONDITION	ANTIBIOTICS?	DETAIL
Localised odontogenic infection (periapical, periodontal or pericoronal abscess)	**No**	Not routinely — if a procedure can be done within 24 hours. Drainage removes the source and the bacteraemia resolves fast. [p131]
Pericoronitis (localised)	**No**	Irrigate with sterile saline, rinse with warm saline or chlorhexidine. Extraction is the treatment of choice; consider removing/recontouring the opposing tooth. Irrigation alone may resolve mild cases. [p130–131]
Spreading odontogenic infection	**Yes**	As an adjunct to the procedure. This is the main indication. [p128]
Necrotising gingivitis (ANUG)	**Yes**	One of the few clear yeses. Debridement + irrigation + antibiotics + analgesia + risk factors (smoking). **metronidazole 400 mg orally 12-hourly for 3–5 days**, or amoxicillin 500 mg 8-hourly if metronidazole unsuitable. [p119–120]
Periodontitis	**Rarely**	Only if rapidly progressing, not responding to treatment, or in immune compromise including suboptimally managed diabetes — and preferably under a periodontist. [p118]
Peri-implant mucositis	**No**	Debridement, hygiene, smoking management, review. Not required. [p122]
Peri-implantitis	**Limited**	Evidence is limited; consider specialist referral before prescribing. If used, alongside other interventions: amoxicillin 500 mg 8-hourly + metronidazole 400 mg 12-hourly, both for 7 days. [p123]
Alveolar osteitis (dry socket)	**Never**	"Antibiotic therapy has no place in the management of alveolar osteitis." It is failed healing, not infection — and it is commonly misdiagnosed as infection. Irrigate with warm saline, long-acting LA, analgesia, obtundent dressing. [p90]
Replanted avulsed tooth	**Consider**	Consider after replanting and splinting: amoxicillin 500 mg (child: 15 mg/kg up to 500 mg) orally, 8-hourly for 7 days. Amoxicillin preferred on clinical experience. Check tetanus status. [p234]

CONDITION	ANTIBIOTICS?	DETAIL
Maxillary sinusitis presenting as dental pain	**Rarely**	Pain worse on bending forward. Symptomatic therapy; antibiotics rarely needed. Dental treatment not required. [p194]
Post-op pain and swelling after oral surgery	**No**	Normal swelling peaks at up to 3 days. Suspect infection only if pain and swelling **worsen after day 4**, or there is purulent discharge, foul taste, progressive trismus, fluctuant swelling or fever. [p89]

7. The traps, collected

If you are skimming this the night before, skim this bit.

- **Antibiotics are not a substitute for dental treatment.** Say it in the station. It is the single most-repeated sentence in this chapter of TG.
- **The 24-hour rule.** Localised infection + procedure within 24 hours = no antibiotic. It is the delay that creates the indication, not the pus.
- **Non-infected retained fragment = no antibiotic.** Infected and delayed = yes.
- **Dry socket is not an infection.** No antibiotics to treat it, and none to prevent it.
- **Joint replacement changes nothing.** Nor do breast implants, shunts or deep brain stimulators.
- **Clindamycin is out for endocarditis prophylaxis.** Still in for severe penicillin allergy in spreading odontogenic infection.
- **Rheumatic heart disease now counts for everyone**, not just Aboriginal and Torres Strait Islander peoples.
- **Metronidazole alone is never right** for odontogenic infection — but it is the drug of *choice* for necrotising gingivitis.
- **Phenoxymethylpenicillin beats amoxicillin** when a procedure is prompt — narrower spectrum.
- **Verify the penicillin allergy** before you route around it.
- **Take the sample before the antibiotic.** Aspirate pus with a needle and syringe into a sterile jar; swabs do not preserve anaerobes and pick up oral flora.

The stewardship sentence, for when an examiner asks why not just prescribe: every unnecessary course carries real harms — diarrhoea, hypersensitivity, and resistance that outlasts the patient in front of you. TG's whole framing is that the narrowest effective drug, for the shortest effective time, only when there is a clear indication, *is* the standard of care — not a cost-saving compromise.

Everything here is sourced from the TG Oral and Dental corpus that powers the TG4 Bible chatbot on this site — page numbers refer to that document, so you can go and read the original wording for anything you want to check. Guidelines change; the endocarditis and clindamycin advice above has already moved once. Treat this as a study aid for the ADC practical exam, not as a prescribing reference for real patients.